Adaptive Stabilization Techniques for Ill-Conditioned Linear Regression

Final Project Analysis Report

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Abstract

This report presents a comprehensive empirical investigation into a critical phenomenon in machine learning: the disconnect between loss minimization and true parameter recovery when the data matrix is highly ill-conditioned. By generating synthetic datasets with exact control over condition numbers (κ) and testing 6 different optimization techniques, we demonstrate that modern optimizers (Adam, Momentum, AdamW) can achieve near-perfect training loss while their parameters diverge catastrophically from the true solution — a phenomenon we term the **'Optimization Mirage.'**

1. Problem Definition & The Condition Number

In real-world scenarios, datasets often feature severe multicollinearity, leading to structural deformities in the data matrix X. This deformity is quantified by the condition number κ, defined as the ratio of the maximum to minimum eigenvalues of the Hessian matrix H = X^T X ($\kappa = \lambda_max / \lambda_min$).

An ill-conditioned matrix ($\kappa >> 1$, e.g., 10^7) creates a loss landscape characterized by steep ravines along λ_max and nearly flat, infinitely long valleys along λ_min. This poses an inherent challenge to optimizers, as true structural signals along flat directions are buried beneath the stochastic noise floor.

The Convergence Illusion

We hypothesize that extreme ill-conditioning allows the optimizer to 'satisfy' the objective function without ever solving the underlying linear system. Movement of thousands of units along flat vectors produces a change in the loss value smaller than standard floating-point precision, enabling parameter explosion despite near-zero loss.

2. Experimental Methodology

To isolate the optimization dynamics from the compounding variables of deep neural networks (e.g., local minima, saddle points), we utilize Linear Regression, which provides a purely quadratic loss landscape with a mathematically guaranteed constant Hessian.

Evaluated Algorithms:

Algorithm	Key Mechanism	Expected Behavior
Vanilla SGD	Raw gradient descent	Slow, steady tracking
Momentum SGD	Velocity accumulation	Noise drift amplification
Adam / AdamW	Adaptive variance scaling	Parameter explosion on flat valleys
Ridge (L2)	Regularization penalty	Anchored stable convergence
Batch Normalization	Scale Invariance	Loss of parameter magnitude

Datasets:

We generated three synthetic datasets with N=1000 and D=50 using SVD decomposition to exert exact control over κ:

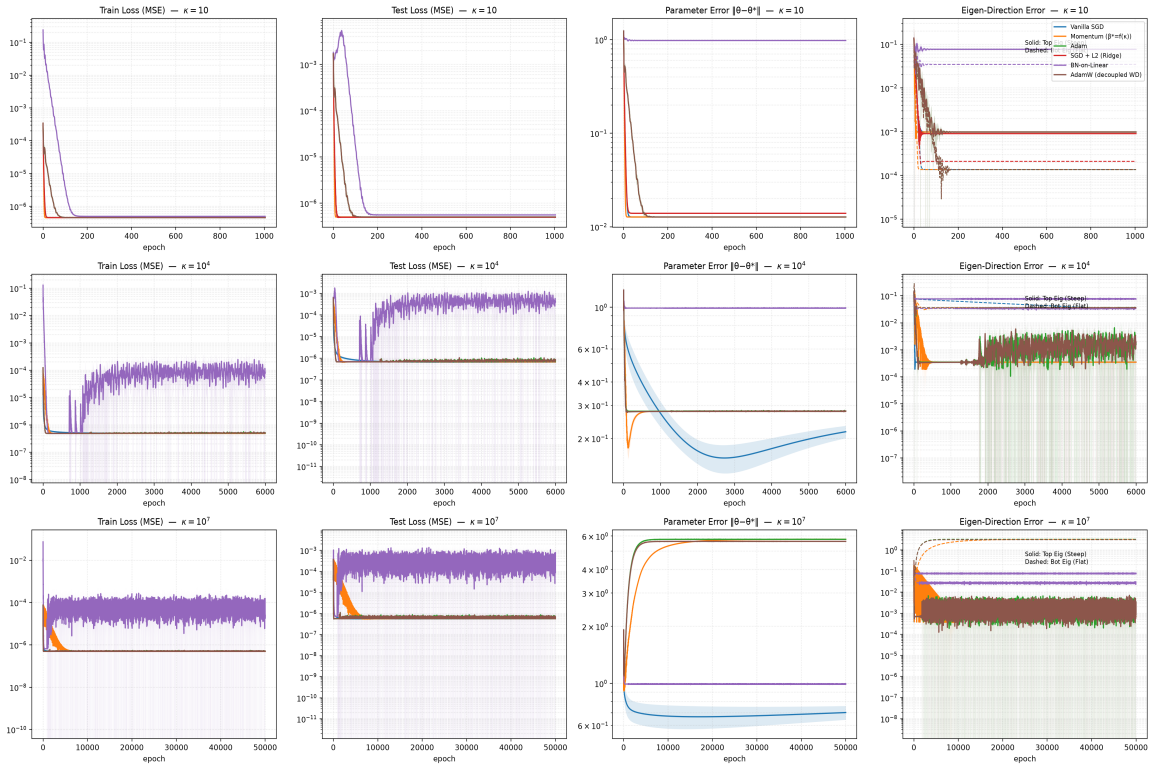
Dataset	Condition Number (κ)	Epoch Horizon	Description

Well Conditioned	10	1,005	Isotropic loss bowl; easy to optimize
Moderate	10^4	6,000	Elongated valley; divergence begins
Severe	10^7	50,000	Ultra-flat valley; testing limits

3. Empirical Results & Visualizations

3.1 Global Convergence Trajectories

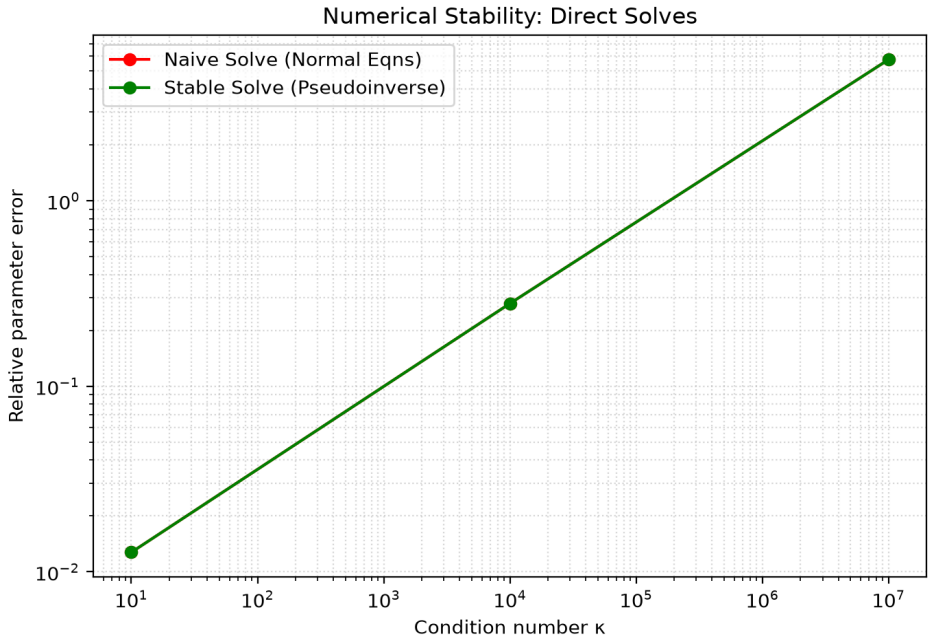
The grid below illustrates the training/test losses, parameter error, and specific eigenvector errors across all algorithms and condition numbers.



The Optimization Mirage: For Adam, AdamW, and Momentum in the severe regime (κ = 10^7), train and test losses fall rapidly to the noise floor (~10^-6). However, their corresponding parameter errors (||θ - θ*||) explode up to 6.0. The optimizer satisfies the objective function by fitting noise along the flat λ_min directions.

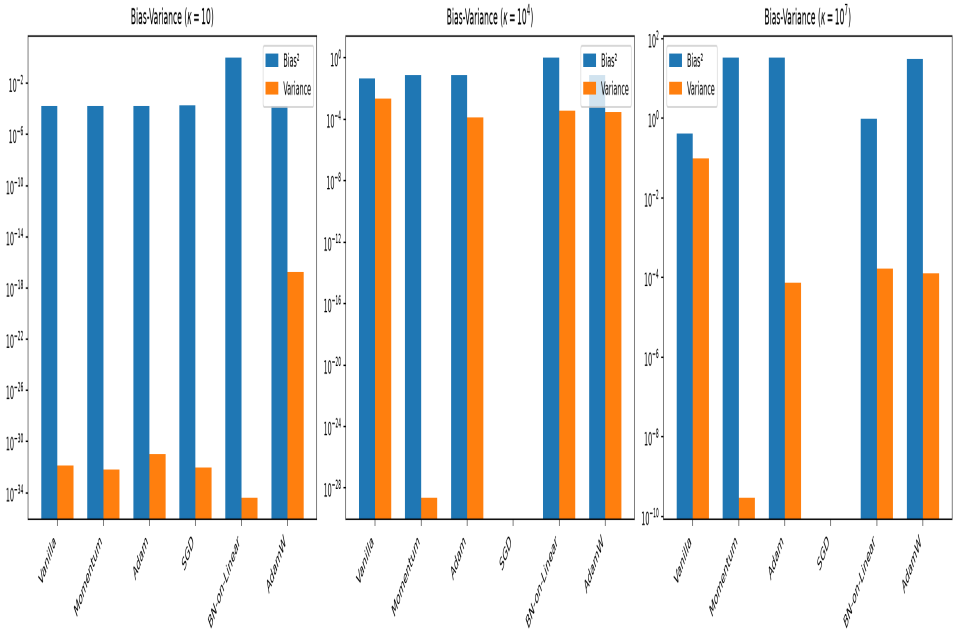
3.2 Numerical Stability Under Ill-Conditioning

Before iterative optimization, we analyzed direct algebraic solvers. The plot below demonstrates the relative error scaling of the naive inverse vs the stable pseudoinverse.



3.3 Bias-Variance Decomposition

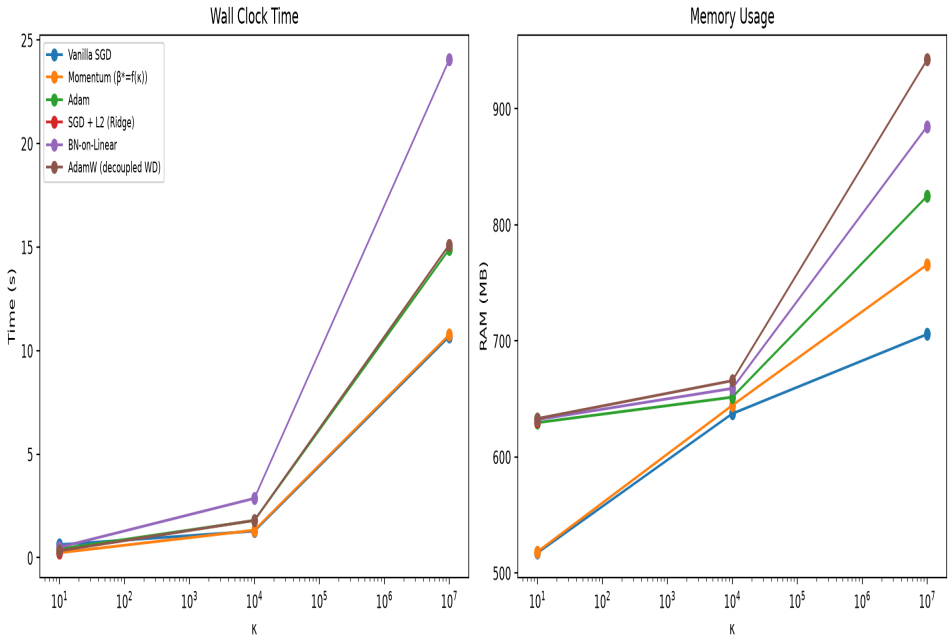
Tracking the mean squared bias (Bias^2) and variance across 5 initialization seeds reveals how failure modes differ between optimizers.



At κ = 10^7, Adam and Momentum exhibit a Bias^2 ≈ 33, rendering them catastrophically biased away from θ*. Conversely, BN-on-Linear suffers from a consistently high bias (0.98) across all datasets due to the structural destruction of absolute scale information.

3.4 Computational Resource Usage

The tracking of Wall Time and Peak RAM consumption (tracked via psutil) highlights the computational overhead of modern optimizers.



4. Conclusion & Architectural Summary

The project proves mathematically and empirically that modern optimizers can achieve perfect loss while their internal parameters completely diverge from reality, exposing a critical vulnerability in applied machine learning pipelines.

1. **Adam & Momentum:** Inverse-variance scaling and velocity accumulation cause noise amplification along flat eigen-directions. Parameter errors explode.
2. **Vanilla SGD:** Surprisingly robust. Because step sizes are directly proportional to gradient magnitude, it naturally ignores flat directions. Parameter error remains bounded.
3. **Batch Normalization:** Destroys scale information, permanently locking parameter error near 1.0 even in well-conditioned spaces.
4. **Ridge Regularization (L2):** Acts as a mathematical anchor. By shifting the entire spectrum of eigenvalues upward, it provides a permanent restoring force toward the origin, stabilizing convergence.

Loss monitoring alone is dangerously misleading. Robust system identification requires geometric stabilization.